

Response to the Reviewers

Vaccination Policy under Model Uncertainty:

Can the Needs of the Few Outweigh the Needs of the Many?

We are sincerely grateful to both reviewers for the care and the generosity of their reports. Reading them, we realised that several of the points raised were not small matters of presentation but places where the manuscript was either imprecise or simply wrong, and we have been glad of the chance to put them right. We are particularly grateful for the references we were pointed to, and for a set of historical counter-examples that sharpened our thinking about the central assumption of the paper.

Below we take each comment in turn. For every comment we give the reviewer’s words in full, then our reply, then the specific changes made to the manuscript. All section numbers refer to the revised manuscript, so that each item can be located directly.

Summary of the changes

Comment	Point raised	Revised ms.
R1 · C1	Engage the selective-lockdown ethics literature	§2.4
R1 · C2	Engage the modelling-humility and uncertainty-communication literature	§5
R1 · C3	Ground the confidence claim in the Simplifications section	§2.5
R1 · C4	Clarify what is negative about “negative herd immunity”	§3.3
R1 · C5	Specify which notion of herd immunity is in use	§3.3
R1 · C6	Introduce the block quotations properly	§1, §2.1
R2 · C1	The virulence-decline premise	§2.1, §3.2
R2 · C2	The claim about tetanus in the introduction	§1

Reviewer 1

Overall comment

Comment.

This is a very good paper which proposes an alternative model for infectious disease spread under conditions of uncertainty. The proposed model yields conclusions and supports recommendations substantially different from those normally derived from models, especially those used during the pandemic. By proposing a plausible model (that is, one based on plausible, albeit questionable, assumptions about disease spread, human behaviour, virulence changes etc) and suggesting that selective, rather than population-wide, vaccination could result in fewer deaths given certain assumptions, the paper emphasises the uncertainty and speculative nature of many epidemiological models used for policy recommendation, thus making a case for (what I take to be) much needed epistemic humility in model-based epidemiology and subsequent policy recommendations.

The paper is clearly written and well structured and I think it would make a nice contribution to the literature on pandemic/epidemic preparedness and on model-based policy making.

For these reasons, I recommend that it is published, though I have some suggestions for improvements and I should point out that my expertise is in ethics/philosophy only. I cannot therefore assess the technical aspects of the proposed model.

Response.

We thank the reviewer warmly for this generous assessment and for the recommendation to publish. We are pleased that the argumentative role of the model came across as intended: the paper is not a prediction about any particular pandemic, but a constructive counter-example meant to show how much weight contestable modelling assumptions can carry. We have adopted all six suggestions below. Four of them (C1, C2, C3 and C6) are straightforward additions or clarifications; C4 led us to rewrite the passage the reviewer found unclear; C5 raises a question about the notion of herd immunity that we address separately.

Comment 1: Engage the selective-lockdown ethics literature

Comment.

About the ethics of selective vaccination, the authors rightly say that additional normative premises are needed to draw ethical implications and policy recommendation from a model predicting fewer deaths with selective vaccination. The literature on selective lockdowns, which were defended ethically on the basis of similar considerations (creating immunity among the non-vulnerable by letting them free to live their normal lives while temporarily shielding the elderly and vulnerable) might offer some insights that the authors might want to engage with or at least refer to. See e.g. Savulescu J, Cameron J. Why lockdown of the elderly is not ageist and why levelling down equality is wrong. *Journal of Medical Ethics* 2020;46:717-721.

Response.

We are grateful for this pointer. The parallel the reviewer draws is close: shielding the vulnerable while allowing immunity to build among the non-vulnerable is structurally very near to our Strategy 2, and the normative premises that were used to defend selective lockdowns are of the kind our paper says are needed here. We have added a reference to that literature at the point where the manuscript states that such further premises are required. We say more about the extent of our engagement in our reply to Comment 2.

Changes.

In §2.4 (*Limits of Ethical Inference from the Model*), the closing sentence of the third paragraph now ends: "... require additional normative premises that lie outside its scope, such claims are, for instance, discussed in the literature on selective lockdowns", with references to Savulescu and Cameron (2020), Cameron et al. (2021) and Giubilini et al. (2021).

Comment 2: Engage the modelling-humility and uncertainty-communication literature

Comment.

About the uncertainty around models, their susceptibility to contested assumption, and the importance of epistemic humility that these factors, many of the recommendations put forward by the authors in the conclusion align with the recommendations in Saltelli A, et al 2020. Five ways to ensure that models serve society: a manifesto. *Nature*. Jun;582(7813):482-484. The authors might want to engage with or refer to those. And about the importance of communicating uncertainty around public health policy, the authors might want to see Giubilini, A et al 2025, “Expertise, Disagreement, and Trust in Vaccine Science and Policy. The Importance of Transparency in a World of Experts”. *Diametros* 22 (82): 7-27.

Response.

Many thanks for drawing our attention to these papers, which are indeed close to our conclusions. Both are now cited. We should be candid about how far we have taken this and the previous suggestion. A proper engagement with either body of work would deserve several pages, and we judged that it would pull the manuscript away from its own line of argument. We have therefore opted for citation rather than discussion, and we would of course be glad to expand if the reviewer thinks the paper would be better for it.

Changes.

In §5 (*Conclusions*), Saltelli et al. (2020) is now cited at the end of the paragraph on uncertainty about what a proper epidemiological model should look like. Giubilini et al. (2025) has been added to the references supporting the recommendation of good uncertainty communication in the penultimate paragraph.

Comment 3: Ground the confidence claim in the Simplifications section

Comment.

The section on Simplifications of the model might benefit from further clarifications. In particular, the authors write at the end of that section that “Despite all these simplifications, we are confident that the main qualitative conclusions our model licences can also be drawn from modified models and/or evaluation metrics”. Where does that confidence come from? Why would the reader trust your confidence? Surely, appeal to your own sense of confidence by itself does not provide sufficient support.

Response.

This is a fair hit, and we accept it. An appeal to our own confidence does not, on its own, provide support, and the sentence as written asked the reader to take our word for something we had not argued. We have added the reasons behind the claim and, equally importantly, we now say plainly how limited the support those reasons provide is. The honest position is this: we did not run the further simulations, so we cannot report them. What we can offer is the heuristic that small changes to a system rarely revolutionise its behaviour, and the observation that more complex simulations tend to be less predictable rather than more, which if anything makes the choice between vaccination strategies harder rather than easier. We prefer to state that and leave readers to draw their own conclusions.

Changes.

In §2.5 (*Simplifications*), the closing paragraph has been extended. After the sentence the reviewer quotes, it now continues: “Since we did not carry out further simulations, we cannot report them and have to rely on heuristics for predicting their results. Our confidence is partly supported by robustness heuristics that small changes to a system rarely revolutionise system behaviour. Furthermore, more complex simulations normally lead to less predictable outcomes. Hence, complicating our model is, in our minds, likely to make it even harder to predict which vaccination strategy is preferable. Ultimately,

we invite readers to draw their own inferences and conclusions; these may be supported by further modelling efforts.”

Comment 4: Clarify what is negative about “negative herd immunity”

Comment.

The notion of ‘negative herd immunity’ the authors introduce at p. 14 might need to be better explained. The authors write “This situation can be understood as a sort of negative herd immunity effect, where herd immunity may have negative consequences for those who remain susceptible”. It’s not very clear why the implications are negative. I suppose herd immunity is good in any case for those susceptible as, by definition, it means a significantly reduced risk of infection (compared to a situation of non-herd immunity). It is not clear to me that ‘herd immunity’ itself is what is negative, as opposed to the way in which it is achieved. If I understand well, on this model, vaccine-induced herd immunity would be less good than collective immunity obtained by allowing natural infection among the non-vulnerable, as the latter is more likely than the former to reduce virulence. But that simply means that the latter is preferable to the former (though neither is negative) and that what is more or less preferable is the way in which herd immunity is achieved, rather than the outcome itself.

Response.

We thank the reviewer for pressing on this, and we accept the reconstruction offered. It is not herd immunity that is bad in our model. What matters is the route by which collective immunity is reached, and what the model tracks is a change of sign over time rather than a bad outcome. Reducing transmission early does protect the population, but in our model it also keeps the virus circulating in a smaller and better-protected pool, so that it ages more slowly. Over the full run of an epidemic, the early benefit can be outweighed, and more agents die. The wording we had borrowed from Luyten et al. (2011) obscured this, since their negative herd immunity effect names a different

mechanism, the shifting of severe illness to a later age. We have rewritten the passage so that the temporal structure of the effect is explicit and so that our use of the term is clearly distinguished from theirs.

Changes.

In §3.3 (*Interpretation*), the clause the reviewer quotes, “where herd immunity may have negative consequences for those who remain susceptible”, has been deleted. The sentence now reads: “This situation can be understood as a sort of negative herd immunity effect, an initial reduction of the transmission rate grants an initial positive immunity effect to the population (herd immunity) that, as time progresses, turns into a negative overall effect causing more agents to die.” The contrast with the age-related mechanism of Luyten et al. is retained at the end of the same paragraph. **To be honest I would not use the expression ‘negative heard immunity’, I think its more confusing than clarifying.**

Comment 5: Specify which notion of herd immunity is in use

Comment.

Moreover, ‘herd immunity’ can mean different things (from a dynamic equilibrium that allows for periodical waves of the virus, as per Sunetra Gupta’s understanding, to simply very low risk that a susceptible and an infectious person come into contact). It would be good to clarify which understanding of herd immunity is used here, as that seems relevant to an epidemiological assessment of the model.

Response.

We are grateful for this question, which turned out to be more than a matter of terminology. The reviewer is right that the choice bears on an epidemiological assessment of the model, and the manuscript had not made it.

The answer is settled by the structure of the model rather than chosen by us. Our SISD model has no removed compartment. Agents who recover return to the susceptible pool carrying partial immunity, which lowers their probability of reinfection and of death without ever eliminating either. Since the susceptible pool is continuously replen-

ished and immunity is never sterilising, there is no threshold whose crossing would bring transmission to an end. What the population reaches is the first of the two notions the reviewer distinguishes: a dynamic equilibrium at which the pathogen remains endemic. The manuscript in fact already displayed this without saying so, in the example run of the appendix, where the epidemic settles into an endemic but relatively harmless state. We have now stated it where the reviewer looked for it.

One divergence from the models usually associated with that notion is worth recording. There, recurrent waves are sustained by antigenic escape, as new variants restore virulence and evade existing immunity. In our model the pathogen instead attenuates with every transmission, so the equilibrium it approaches is not merely endemic but progressively benign. We have made this explicit as well, so that our appeal to a dynamic equilibrium is not read as a claim about antigenic drift.

Changes.

In §3.3 (*Interpretation*), a new paragraph now precedes the discussion of the negative herd immunity effect: “Herd immunity is used in more than one sense, and our model settles which one applies. There is no removed compartment: agents who recover return to the susceptible pool carrying partial immunity, which lowers their risk of reinfection without ever eliminating it. No threshold is crossed after which transmission ceases. The population instead reaches a dynamic equilibrium in which the pathogen remains endemic at a low level, as the example run in Figure 6 illustrates. It is herd immunity in this sense that we appeal to here. Unlike models in which new variants restore virulence, the pathogen in our model attenuates with each transmission, so the equilibrium it approaches is not merely endemic but progressively benign.”

Comment 6: Introduce the block quotations properly

Comment.

Small comment: There are citations that are added to the text without context or being introduced properly. For instance it is unclear where the citation at pp.5-6 comes from and how it relates to the discussion there, and same for citation at p. 8. Citations

should be properly introduced in a way that makes them fit with the flow of the text.

Response.

We fully agree, and we are grateful for the observation. The passages in question are block quotations that we use to support our own reasoning, but we had dropped them into the text without saying whose words they are or why they appear at that point. Each is now introduced by name.

Changes.

In §1 (*Introduction*), the quotation from Brüssow is now preceded by “Brüssow wrote during the pandemic:”. In §2.1 (*Main Components of the Model*), under the heading on viral age, the quotation from Kun et al. is now preceded by “In the words of Kun et al.:”.

Reviewer 2

Overall comment

Comment.

Thank you for asking me to review this paper. The more general points in the paper about the limits of modelling and of assumptions that everyone must be vaccinated are important points.

Response.

We thank the reviewer for the time given to the manuscript, and for granting the general points about the limits of modelling. The central objection that follows is, in our view, the most serious challenge the paper has received, and we take it up in detail below.

Comment 1: The virulence-decline premise

Comment.

However, the main thrust of the paper depends on a doubtful empirical claim, that pandemic viruses decline in virulence. While many people might believe this claim, and

while there might be evidence for effects of this kind, such effects are clearly dwarfed by the effects (acting in the same direction on average disease severity) of acquired immunity, whether via vaccination or (as was more common in historic pandemics) previous infection.

Consider these counter-examples: (1) Hong Kong & omicron - a very high death rate was observed due to a lack of immunity due to low prior infection rates and low vaccination among older adults; in other words, low population immunity was a much bigger factor than any decline in the intrinsic virulence of omicron. (2) remote communities of the pacific or the arctic where epidemics of 1918 flu were typically late but nonetheless severe. More generally (3) the high mortality from smallpox and measles when introduced to immuno-naive populations via colonisation, suggesting again that a lack of prior immunity was a much bigger factor than decline in virulence from centuries of endemic transmission in Europe.

These difficulties with the core assumption make it difficult to evaluate any ethical implications that follow.

Response.

We are grateful for this objection. It is the most serious challenge the manuscript received, and the three counter-examples are well chosen.

We should say first what we are not disputing. We do not claim that virulence decline was the dominant factor in the cases the reviewer names, and we accept that in each of them low population immunity was the larger effect. We also accept that the magnitude of intrinsic virulence decline in real pandemics is contested. The manuscript now says so.

What our argument requires is weaker than what the objection denies. The claim we examine is that mass vaccination would surely improve public health. That is a claim of necessity, and a single coherent counter-example is enough to defeat it. Our model is offered as a proof of concept in exactly that sense. It does not predict what any pandemic will do. It exhibits a mechanism, places it in a model no less reasonable than others in use, and shows that the ethical verdict can reverse. For this the transmission–virulence trade-off need not be typical, or even common. It needs to be real and of uncertain

magnitude. The reviewer’s own formulation grants the first, allowing that there might be evidence for effects of this kind, and disputes the second. A dispute about magnitude is precisely the kind of uncertainty this paper is about.

This is why the objection, granted in full, does not weaken our conclusion. Suppose virulence decline really is dwarfed by acquired immunity in most pandemics. Then the models that guided policy encoded one particular weighting of two mechanisms whose relative size was contested at the time and remains contested now. A different weighting yields a different verdict about free-riding and about who ought to be vaccinated. We do not claim that our weighting is the right one. We claim that the verdict is sensitive to it, and that this sensitivity is a reason for humility about the moral conclusions drawn from such models. The objection supplies further evidence that the weighting is uncertain, which is our thesis rather than an objection to it.

Two smaller points. Our model represents acquired immunity alongside viral ageing, so we do not attribute to one mechanism what the other does. And our aim is a model that may be useful in future health emergencies rather than a retrodiction of past ones. Whether the trade-off dominated in 1918 or in 2022 is a different question from whether it could matter next time.

Changes.

Two changes prompted by this comment have already been made to the manuscript. In §2.1 (*Main Components of the Model*), a new paragraph follows the description of viral age. It cites all three of the reviewer’s counter-examples, namely the Omicron BA.2 variant in Hong Kong, the isolated Pacific islands after the 1918–1919 influenza pandemic, and immuno-naïve populations encountering pathogens from European colonisers, and it grants that “virulence decline can be small and dwarfed by acquired immunity or even non-existent”. It then sets against this the evidence that immunity to Covid-19 waned over time and fell sharply after Omicron. In §3.2, a sentence now directs the reader to those parameter settings with a small viral age effect that correspond to the cases the reviewer raises.

Comment 2: The claim about tetanus in the introduction

Comment.

The introduction also contains the incorrect claim that tetanus has been virtually eradicated via herd immunity - this is not true, at least not insofar as “herd immunity” means something like reduced population transmission arising from the presence of transmission-reducing immunity in large numbers of individuals, because tetanus is not transmitted between people.

Response.

The reviewer is right, and we are grateful for the correction. Tetanus is contracted from environmental spores and is not transmitted between people, so it cannot be controlled by herd immunity and should not have been offered as an example of it. The opening sentence has been corrected.

Changes.

In §1 (*Introduction*), the opening sentence previously read “... such as the eradication of smallpox and the virtual eradication of tetanus and polio by achieving herd immunity”. It now reads “... such as the eradication of smallpox and the virtual eradication of tetanus (among vaccinated patients) and polio”. The appeal to herd immunity has been dropped from the sentence.

Further changes

Beyond the points raised by the reviewers, we have made two editorial passes over the manuscript. Em dashes have been replaced by parentheses throughout, and multiple citations within a single bracket have been ordered alphabetically. In §3.5 (*Worst Cases*) we have also added a short remark that decision makers are likely to be reluctant to bear the costs should selective vaccination backfire, which makes worst-case outcomes of practical as well as theoretical interest.

We thank both reviewers again for their attention to the manuscript, which we believe is considerably stronger for it.